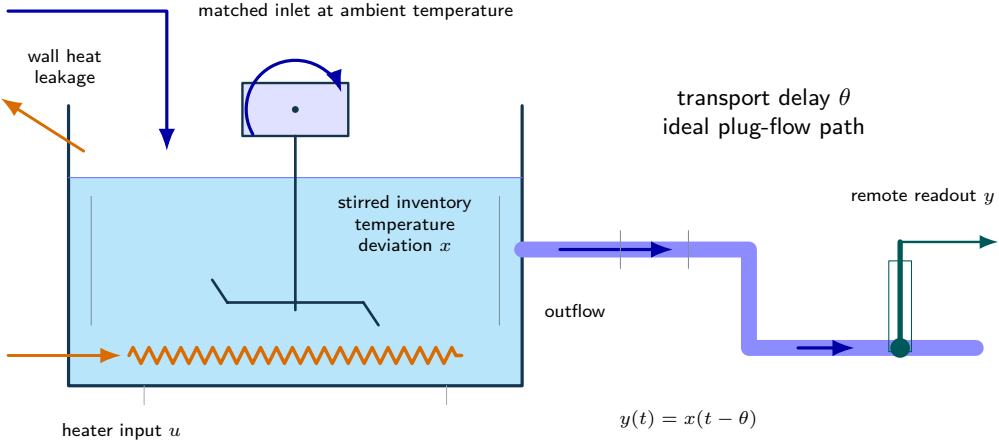

One physical interpretation of a first-order-plus-delay model

Synthetic experiment

$G(s) = 1.8e^{-0.6s}/(4s + 1)$. Declared gain, lag and delay generate the records; none is a physical measurement.

Thermal interpretation

One mixed energy store plus ideal transport. Through-flow and wall losses enter one effective conductance.

Identification boundary

Fit on training data, freeze the model, then check a separate multistep record. Geometry does not identify capacity or heater efficiency.